

Displacement-based seismic design of buildings—theory

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Abstract

The conceptual basis of the spectral acceleration-based design method currently used in seismic codes is reviewed and its limitations are discussed. An alternative method that uses displacements as the basis for the design procedure is then presented. Its conceptual basis for elastic and inelastic seismic design and its application to single-degree-of-freedom and multi-degree-of-freedom structures are reviewed. The effects of torsion are considered. The advantages of this method over the spectral acceleration-based design method are also discussed. To the authors' knowledge, this and the accompanying paper mark the first application of the displacement-based procedure to the seismic design of (steel) buildings. © 1999 Elsevier Science Ltd. All rights reserved.

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1. Introduction

The National Building Code of Canada (NBCC) [1] and other design codes [2] use a spectral acceleration-based method for seismic design. Consider a structure that may be represented by a single-degree-of-freedom (SDOF) system of mass, m , elastic period, T_e , and damping ratio, ξ . The period, T_e , is estimated from empirical equations derived from experiments on structures principally designed for high seismic zones. Fig. 1 shows the

normalized design acceleration response spectra, S_a/g , specified by the NBCC [1] for zones where the zonal acceleration and velocity are numerically equal. The elastic spectral acceleration, $(S_a/g)_e$, corresponds to the estimated period, T_e . The elastic base shear that a SDOF system of mass M attracts is

$$V_e = Mg(S_a/g)_e \quad (1)$$

If the SDOF system is provided with strength V_i that

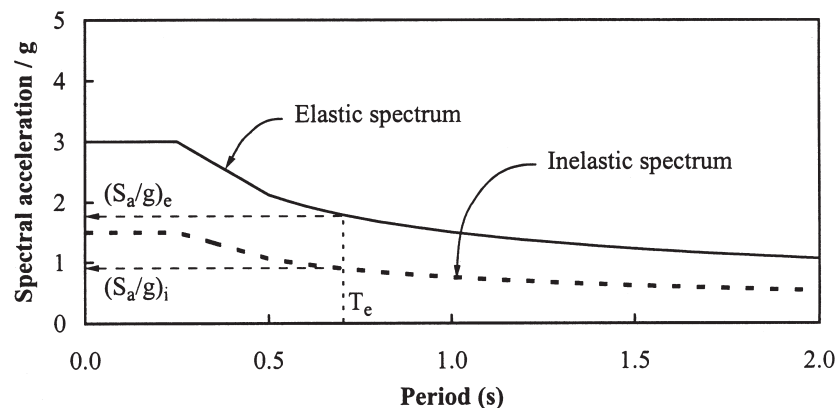


Fig. 1. Design acceleration response spectra.

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is less than V_e but has sufficient ductility, it responds inelastically as shown in Fig. 2 (assuming that the initial period remains equal to T_e). The NBCC [1] accommodates this behaviour by reducing the elastic spectrum by the force modification factor, R , to arrive at the inelastic design spectrum shown in Fig. 1. The inelastic base shear is thus

$$V_i = \left(\frac{V_e}{R} \right) = Mg(S_a/g)_i \quad (2)$$

Inelastic seismic design therefore achieves economy because the system is designed for a reduced base shear of V_i rather than V_e . However, ductility is required to ensure that this inelastic response is obtained and, as well, it is implicit that the structural and non-structural damage consistent with the response must be accepted.

The spectral acceleration-based design method has the following limitations.

1. An estimate of the fundamental period, T_e , is required to start the design process because the period of the structure to be designed is not known. Seismic codes use empirical expressions for the period based on a general description of the structural system and its geometry. These estimated periods are intentionally less than the measured ones to give a conservative design [3,4].
2. The force modification factor, R , is intended to be a simple means of arriving at an inelastic design. Seismic codes specify values for the factor, R , depending upon the material of construction and the type of structural system used. However, these values appear to be arbitrary, are difficult to justify, and do not appear to have been established consistently by experiment or analysis.
3. Displacements are treated in a somewhat cursory manner and are checked at the end of the design process only. There appears to be a lack of concern about the implied inelastic displacements when values of R greater than 1.0 are used. Both non-structural and structural elements may be deemed unsatisfactory if they deform excessively under earthquakes associated

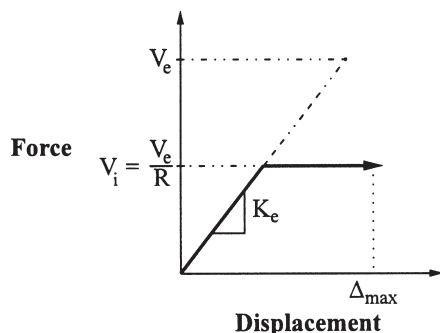


Fig. 2. Inelastic response of a single-degree-of-freedom system.

with the serviceability limit state. At the ultimate limit states, the deformations are likely to contribute to the instability of the structure and, as well, the damage, perhaps implying that the building is partially or completely non-functional or even beyond repair, may also be considered unacceptable. These limit states are governed by deformations. Furthermore, it may be easier to define failure of a structural element as a limit on deformation rather than as a limit on the force.

Therefore, it seems rational to examine a seismic design method wherein displacements are considered at the start of the design process with attention focused on deformations to provide a structure that meets the requirements for the several limit states. A serviceability limit state on deformations could be applied under moderate earthquakes that are likely to occur relatively frequently in the life of the structure (by imposing drift limits so that non-structural damage is limited or does not occur). To prevent collapse in a major earthquake, the ductility demand on the structural elements and the overall deformation of the structure would have to be controlled. It is suggested this can be achieved more rationally with a displacement-based rather than an acceleration-based design method.

Therefore, the objectives of this paper are: (a) to present the theoretical basis of the displacement-based design method, and (b) to show how this method is applied, as may be used for the serviceability limit state and the ultimate limit states, to the design of buildings modelled as multi-degree-of-freedom systems. The method is applied to the design of concentrically braced frames (CBFs) in the companion paper [5].

2. Displacement-based design method for a SDOF system

The central concept of the method, based on Priestley [6], is that a structure is designed for a specified target displacement. The method is illustrated by reference to a single-storey, single-bay CBF that is modelled as a SDOF system as shown in Fig. 3. A set of elastic dis-

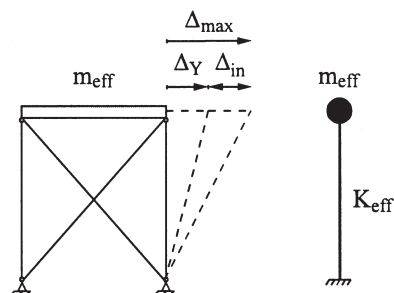


Fig. 3. Single-degree-of-freedom model of a concentrically braced frame.

placement spectra for different levels of equivalent viscous damping is required, as shown in Fig. 4. The set is generated by integrating numerically the equation of motion of an elastic SDOF system subjected to earthquake ground motions appropriate to the location of the structure. The design procedure for a SDOF system is as follows:

1. Estimate the yield displacement of the structure, Δ_Y . For the CBF shown in Fig. 3, this displacement is the lateral drift that yields the brace and, if column deformation is neglected or is minimal, can be taken to be a function of the geometry and material properties (E , F_Y) of the brace only.
2. Select an appropriate maximum inelastic displacement, Δ_{in} , which depends on the deformation capacity of the structural elements. For a CBF with nominal ductility, Δ_{in} may be taken as $1.0 \Delta_Y$, i.e. the ductility demand is 2.0. A larger value may be selected for appropriately detailed structures.
3. The maximum displacement of the SDOF system, Δ_{max} , the sum of the yield displacement, Δ_Y , and the maximum inelastic displacement, Δ_{in} , is thus related to the brace ductility demand assumed in step 1.
4. Select an appropriate value of effective structural damping, ξ_{eff} , which depends on the ductility level implied in step 2.
5. The effective period, T_{eff} , corresponding to the maximum displacement, Δ_{max} , and the effective damping, ξ_{eff} , is obtained as shown in Fig. 4.
6. The effective stiffness of the SDOF system, K_{eff} , is

$$K_{eff} = \frac{4\pi^2 m_{eff}}{T_{eff}^2} \quad (3)$$

where m_{eff} is the mass of the SDOF system.

7. The base shear capacity required, as shown in Fig. 5, is

$$V_b = K_{eff} \Delta_{max} \quad (4)$$

where K_{eff} is the secant stiffness when the structure behaves inelastically, as discussed in the next section.

8. Select structural elements to provide the base shear capacity, V_b . Once the members are selected, the initial design of the structure is completed and the structure can now be analyzed. The elastic stiffness and a revised estimate of the yield displacement are obtained.
9. Revision is necessary when the members selected in step 8 provide significantly different stiffness and strength from that required. The displaced shape and effective damping may be revised and steps 5 to 8 are repeated until a satisfactory solution is obtained.

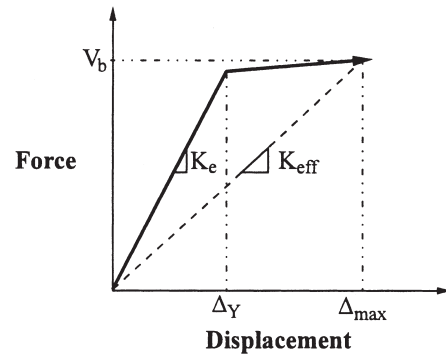


Fig. 5. Elastic and effective stiffnesses.

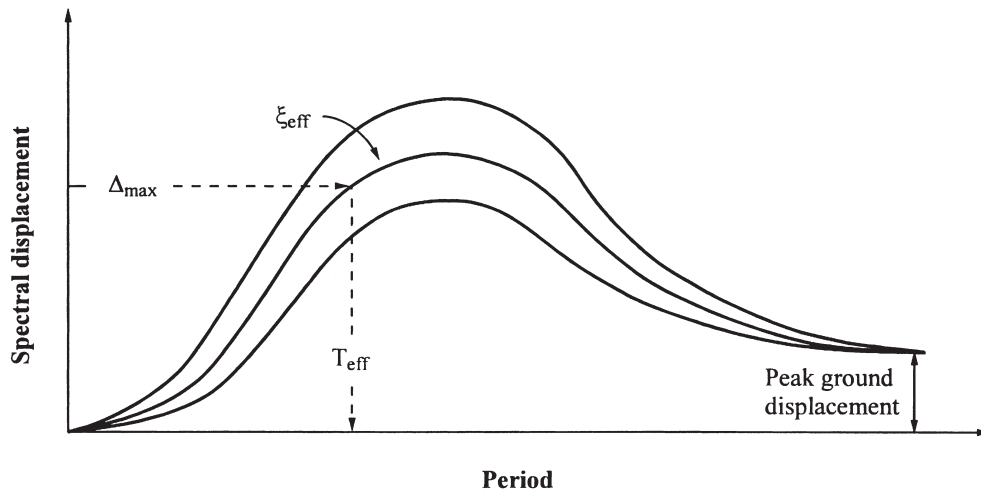


Fig. 4. Elastic design displacement spectra.

3. Basis of the displacement-based design method

When a structure behaves inelastically in the design earthquake, the substitute structure approach is followed in order that elastic displacement spectra can be used in the displacement-based design method. In this procedure [7], an inelastic SDOF system is modelled as an equivalent linear elastic analogue having the ‘substitute’ properties of: (a) effective stiffness, K_{eff} , as shown in Fig. 5, (b) effective damping, ξ_{eff} , and (c) effective period, T_{eff} . The maximum displacement of the inelastic SDOF system is estimated by using the substitute properties together with an elastic displacement spectrum.

Consider the response of an inelastic SDOF system with an elastic period, T_e , and damping ratio, ξ . Let Δ_Y be the yield displacement and Δ_{max} be the maximum target displacement. Considering Fig. 6 where displacement spectra for increasing damping ratios ξ are shown, there are several paths to reach the maximum displacement, Δ_{max} , starting from the point ‘o’ at T_e and Δ_Y on an elastic displacement spectrum [8]: (i) with the same period, T_e , select a lower damping ratio to reach point ‘a’, (ii) with the same damping ratio, select a longer period, T_1 , to reach point ‘b’, (iii) select both a longer period, T_2 , and a larger damping ratio to reach point ‘c’, and (iv) select a shorter period, T_3 , and a lower damping ratio to reach point ‘d’. The first and last alternatives are unrealistic and the third is the most realistic because structures responding inelastically exhibit both greater damping and a longer period due to softening. Gulkan and Sozen [7] conducted dynamic tests on SDOF concrete frames and deduced guidelines for selecting the substitute period and damping. The substitute period was determined experimentally by taking the ratio of the maximum displacement to the maximum absolute acceleration and using the relation $S_a = \omega^2 S_d$, where S_a is the spectral acceleration, S_d is the spectral displacement, and ω is the circular frequency. The substitute damping was evaluated by assuming that the energy input to the SDOF system during the earthquake excitation was tot-

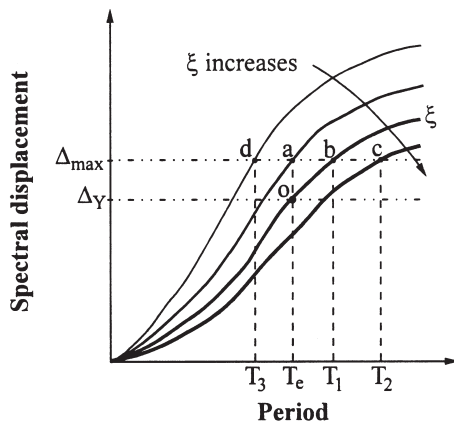


Fig. 6. Use of elastic displacement spectra.

ally dissipated by an equivalent linear viscous dashpot. Therefore, in Fig. 4, the effective damping, ξ_{eff} , is the equivalent viscous damping which comprises damping that exists in the structure without inelastic behaviour (nominal viscous damping) and the hysteretic damping due to inelastic action. The latter, of course, depends on the level of inelastic action in the structure. Shibata and Sozen [9] extended this approach to the design of reinforced concrete frames modelled as multi-degree-of-freedom (MDOF) systems.

For design, the lengthened or substitute period is that corresponding to the assumed maximum displacement, Δ_{max} , and effective damping, ξ_{eff} , that is point ‘c’ in Fig. 6. The damping, ξ_{eff} , depends on the ductility demand imposed on the structural elements in the earthquake. Guidelines for selecting ξ_{eff} for reinforced concrete structures are given by Gulkan and Sozen [7] and Shibata and Sozen [9]. Little information is available on effective damping values for steel structures responding inelastically. Newmark and Hall [10] recommend a damping ratio between 5% and 7% for steel structures with welded connections and between 10% and 15% for steel structures with bolted connections.

Once the substitute period, T_{eff} , is determined for a desired maximum displacement, Δ_{max} , the effective stiffness of the SDOF system, which is a secant stiffness to the maximum displacement, Δ_{max} , as shown in Fig. 5, is found from Eq. (3). [If the period, T_e , corresponding to the yield displacement, Δ_Y , is used in Eq. (3), the elastic stiffness, K_e , is obtained.] The inelastic base shear for the SDOF system is given by Eq. (4).

An additional issue that needs to be addressed is the decrease in base shear with an increased value of the assumed maximum displacement, Δ_{max} . This is illustrated in Fig. 7(a) for the SDOF system shown in Fig. 7(b), where Δ_1 and Δ_2 are two values of the assumed maximum displacement, Δ_{max} , that are both greater than the yield displacement, Δ_Y .

Consider the design of the SDOF system for a maximum displacement equal to Δ_i . The effective period, T_i , is determined from an elastic displacement spectrum corresponding to the assumed maximum displacement, Δ_i , and effective damping, ξ_{eff} . The effective stiffness, K_{eff} , is obtained from Eq. (3) and the inelastic base shear from Eq. (4). Let K_e be the elastic stiffness obtained from Eq. (3) for period, T_e , that corresponds to the yield displacement, Δ_Y . Assuming the mass, m , to remain constant, the ratio of the stiffness, K_{eff} to K_e , is obtained as

$$\frac{(K_{\text{eff}})_i}{K_e} = \left(\frac{T_e}{T_i} \right)^2 \quad (5)$$

where T_i is the period corresponding to the assumed maximum displacement, Δ_i . Fig. 8 shows an idealized design displacement spectrum wherein the spectral dis-

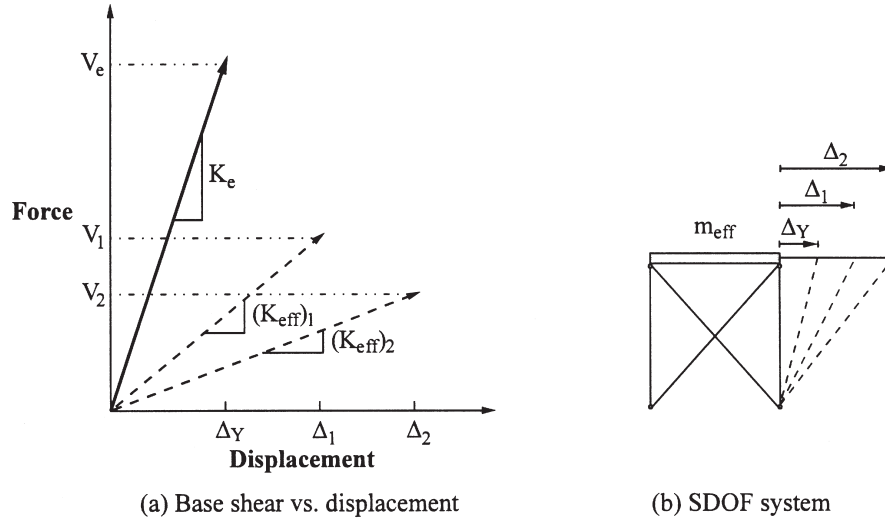


Fig. 7. Reduction in base shear with displacement.

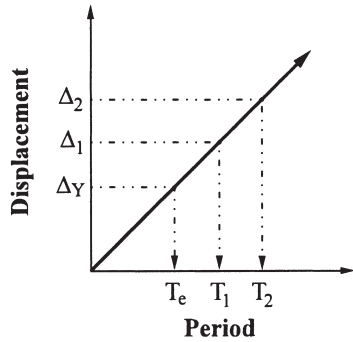


Fig. 8. Linearized design displacement spectrum.

placement is assumed to increase linearly with period. This linearization may be considered acceptable for the range of periods within which the fundamental period of most low-rise to moderate-rise buildings lies. Using the assumed linear relation between period and spectral displacement, Eq. (5) simplifies to

$$\frac{(K_{\text{eff}})_i}{K_e} = \left(\frac{\Delta_Y}{\Delta_i}\right)^2 \quad (6)$$

Using Eq. (4), the ratio of inelastic to elastic base shear is

$$\frac{V_i}{V_e} = \frac{\Delta_Y}{\Delta_i} \quad (7)$$

Thus, for the linearized design displacement spectrum shown in Fig. 8, the base shear decreases in inverse proportion to the ratio of the assumed maximum displacement, Δ_{max} , to the yield displacement, Δ_Y , or conversely the maximum displacement equals the force modification factor, R , times the yield displacement when the spectral displacement varies linearly with the period.

4. Displacement-based design method for a MDOF system

Consider a MDOF system with n degrees of freedom as shown in Fig. 9. The displacement-based design method is applied to this system by first transforming it into an equivalent SDOF system, as outlined by Calvi and Kingsley [11], and then using the procedure outlined in Section 2 for a SDOF system. The MDOF system is forced to behave like a SDOF system by permitting displacements of only a single pre-determined or assumed shape. The transformation is based on the following assumptions:

1. the MDOF system responds harmonically in the assumed shape,
2. the base shear developed by the MDOF system and its SDOF equivalent are the same, and
3. the work done by the lateral earthquake force on both systems is the same.

Let the assumed displacement vector of this MDOF system be represented by

$$\{\delta\} = \{\delta(h,t)\} = \langle \delta_1(t), \delta_2(t), \dots, \delta_n(t) \rangle^T \quad (8)$$

This vector may be expressed as

$$\{\delta(h,t)\} = \{\Phi(h)\}Z(t) \quad (9)$$

where $\Phi(h)$ is a spatial (shape) function and $Z(t)$ is a temporal function. Assuming harmonic response, Eq. (9) is written as

$$\{\delta(h,t)\} = Z_0 \sin(\omega t) \{\Phi(h)\} \quad (10)$$

This expresses the first assumption that the shape of

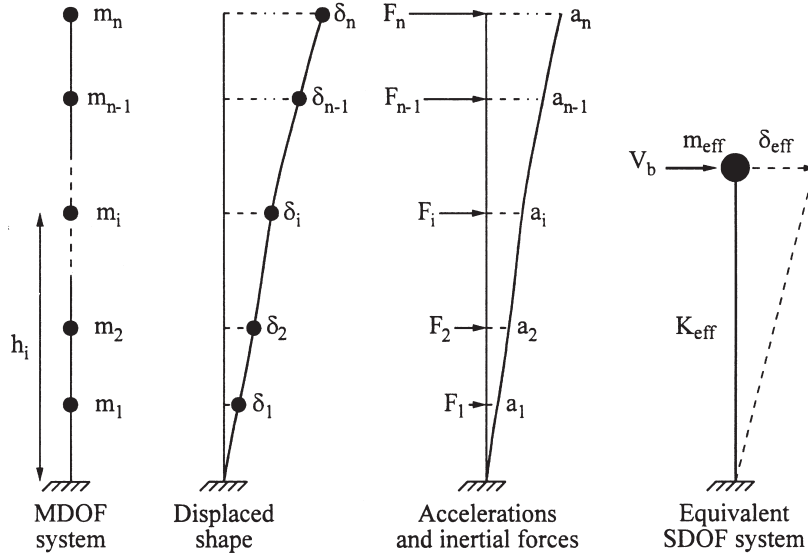


Fig. 9. Multi-degree and equivalent single-degree-of-freedom system.

the MDOF system does not change with time and only the amplitude of motion varies harmonically. Differentiating Eq. (10) twice with respect to time gives the acceleration vector

$$\{a(h,t)\} = -Z_o\omega^2\sin(\omega t)\{\Phi(h)\} = -\omega^2\{\delta(h,t)\} \quad (11)$$

Thus, the acceleration at each DOF, a_i , is proportional to its assumed displacement, δ_i .

Let the properties of the equivalent SDOF system be mass, m_{eff} , stiffness, K_{eff} , damping, ξ_{eff} . Also let δ_{eff} and V_b be the effective displacement and base shear of the equivalent SDOF system, respectively, where these properties are to be derived from those of the MDOF system. The displacement of the MDOF system can be normalized by dividing by the effective displacement of the equivalent SDOF system to give

$$\{c(h,t)\} = \frac{1}{\delta_{\text{eff}}} \{\delta(h,t)\} \quad (12)$$

or

$$c_i = \frac{\delta_i}{\delta_{\text{eff}}} \quad (12a)$$

As accelerations for harmonic motion are proportional to displacements

$$c_i = \frac{a_i}{a_{\text{eff}}} \quad (13)$$

or

$$a_i = c_i a_{\text{eff}} \quad (13a)$$

where a_{eff} is the acceleration of the equivalent SDOF system. Therefore the lateral inertia force at the 'i'th mass is given by

$$F_i = m_i a_i = m_i c_i a_{\text{eff}} \quad (14)$$

By assumption 2, the sum of the lateral forces on the MDOF system is equal to the base shear V_b , thus

$$\begin{aligned} V_b &= \sum_{i=1}^n F_i = \sum_{i=1}^n m_i a_i = \left\{ \sum_{i=1}^n m_i c_i \right\} a_{\text{eff}} \\ &= m_{\text{eff}} a_{\text{eff}} \end{aligned} \quad (15)$$

From Eq. (15), the effective mass of the SDOF system is defined as

$$m_{\text{eff}} = \sum_{i=1}^n m_i c_i \quad (16)$$

The lateral force, F_i , at each mass from Eq. (14) may be expressed in terms of the base shear, V_b , by manipulating Eqs. (12a), (14) and (15) to give

$$F_i = \frac{m_i \delta_i}{\sum_{j=1}^n m_j \delta_j} V_b \quad (17)$$

The effective displacement, δ_{eff} , is found from assumption 3 by equating the work done by the lateral force on each system

$$V_b \delta_{\text{eff}} = \sum_{i=1}^n F_i \delta_i \quad (18)$$

Solving Eq. (18) for δ_{eff} and substituting for F_i from Eq. (17) results in

$$\delta_{\text{eff}} = \frac{\sum_{i=1}^n m_i \delta_i^2}{\sum_{i=1}^n m_i \delta_i} \quad (19)$$

The effective displacement from Eq. (19) is used to enter the displacement spectrum, for the appropriate level of effective damping, to find the effective period. With this period and the effective mass from Eq. (16), the effective stiffness is found from Eq. (3) and the base shear is given by

$$V_b = K_{\text{eff}} \delta_{\text{eff}} \quad (20)$$

The forces at each mass, F_i , are found from Eq. (17) and the design is effected.

5. Design procedure for a MDOF system

The design procedure for a generalised MDOF system is:

1. Select an initial desired displaced shape for the structure, δ_i . For a multi-storey CBF, the desired drift at each floor level is specified. The brace ductility demand can be calculated from the assumed storey drift based on the approximation that it depends only on the geometry and material properties of the frame. (This neglects the axial deformations of the column, which may be appropriate for low-rise structures). Methods for estimating the contributions of column deformations to drift are discussed in the companion paper [5].
2. Select the effective damping, ξ_{eff} , for the structure. This depends on the ductility implied with the desired displaced shape, δ_i .
3. Determine the effective displacement, δ_{eff} , from Eq. (19).
4. Obtain the normalized profile, c_i , from Eq. (12).
5. Obtain the effective mass, m_{eff} , from Eq. (16).
6. Determine the effective period of the SDOF system, T_{eff} , from an elastic design displacement spectrum for displacement, δ_{eff} , and damping, ξ_{eff} .
7. Obtain the effective stiffness, K_{eff} , from Eq. (3).
8. Obtain the base shear, V_b , from Eq. (20).
9. Obtain the lateral forces on the MDOF system from Eq. (17).

10. Design the lateral load resisting system (LLRS) for these forces and the appropriate gravity load. Material design standards are used at this stage to select members and details that have adequate deformation capacity as implied in the desired displaced shape in step 1.
11. With appropriate members selected, a design is now available for analysis. As a minimum, a nonlinear static analysis of the structure under the lateral earthquake force profile is recommended. This analysis accounts for approximately the redistribution of internal forces when the structure is loaded into the inelastic range. The objectives of this analysis are to check the displaced shape against the desired one assumed in step 1 and to assess the strength and deformation demands on the structural elements. In following this analysis procedure, one of two cases may obtain. In the first case, the lateral force profile is applied and the load factor is incremented until the lateral force is attained. Inelastic action may occur at some locations in the structure. (For direct-acting bracing, the use of a small strain-hardening modulus is required so that the deformations are not indeterminate). In the second case, the lateral forces in the assumed profile are applied until the desired deformation is reached at a specified location, e.g., the roof level. Inelastic action does take place in the structure. Obviously, nonlinear dynamic time history analyses give a more realistic assessment of the drift and the ductility demands. When the hysteretic behaviour of components is modelled directly, the effective damping ratio to be used is that pertaining only to other sources of damping (nominal viscous damping).
12. If the displaced shape and the associated deformation demands on the structural elements from the nonlinear analyses are significantly different from those assumed, the stiffness and strength distribution should be revised to obtain the desired result. This is most likely to occur when higher modes of vibration have a significant effect.

6. Torsional effects

Buildings should be designed and constructed, insofar as possible, by modifying the frame stiffnesses to minimise the structural eccentricity, defined as the distance measured perpendicular to the direction of seismic loading between the centre of mass and the centre of rigidity, at the level being considered. Whatever structural eccentricity remains results in a torsional moment and attendant shear forces that are distributed on the basis of equilibrium and frame interstorey drifts to the frames. The adequacy of the frames is checked and they are modified as required.

Paulay [12,13] describes a design strategy for earthquake-induced torsion in buildings which is based on the deformation capacity of the structural system and its critical elements rather than on torsional strength, as is consistent with the displacement-based design approach. When inelastic action occurs, based on what is essentially a lower bound approach for elements displaying elasto-plastic behaviour, Paulay [12] assumes that these elements parallel to the lateral force have yielded due to translation and therefore that the torsional moment can only be resisted by the perpendicular elements whose resistances must form a couple. If a couple cannot be formed, the system is considered to be torsionally unrestrained. For torsionally restrained systems, the elements resisting torsion are likely to behave elastically. In any case, together with the assumption of rigid body rotation of the floors, the other assumptions reduce the system to a statically determinate one. All the frames in the direction of the earthquake are at their yield capacities. The torsional couple formed by the earthquake force acting at an eccentricity measured perpendicular to the earthquake force from the centre of mass to the centre of strength is resisted by the perpendicular elements. The torsional deformations of these elements are taken to be proportional to the distance from the centre of strength (the location where the resultant of the resistances of the inelastic translatory elements acts). Because of the twisting displacements, the centre of mass has an additional displacement above the translational one and some structural or non-structural element becomes the critical one depending on the drift criteria established. Both effects must be considered. Because of the elastic behaviour of the elements resisting the torsion, it is likely that the increase in the displacement of the centre of mass due to torsion is small. Therefore, a logical approach is to base the initial design on the translation of the centre of mass and to modify this assumption subsequently as required. A simple design strategy for torsion therefore exists.

7. Advantages of the displacement-based design method

The displacement-based design method is considered to offer the following advantages over the spectral acceleration-based design method.

1. Displacements play a major role at the preliminary design stage itself resulting in good control on displacements over the entire design process. Target displacement criteria are selected for the serviceability and ultimate limit states and thus damage control is achieved directly.
2. The strength and stiffness of the lateral load resisting system (LLRS) are chosen to satisfy the desired deformation criteria.

3. Empirical equations for estimating the fundamental period of the structure for preliminary design of the LLRS are not required.
4. The selection of a displaced shape at the start of the design process forces the engineer to consider the configuration of the LLRS and the drift tolerance for the non-structural elements. The displaced shape may be linked explicitly to the member ductility demand, as is the case in the CBF. The drift of non-structural elements at various levels of damage can be obtained from experiments and used directly in design.
5. The empirical and somewhat arbitrary force modification factor, R , used in the spectral acceleration-based design method, is not needed.

8. Closure

The displacement-based design method, as outlined in Sections 2 and 4, has been applied to the seismic design of RC bridge columns [14] and MDOF bridge structures [11], respectively. To the authors' knowledge, this method has not been applied to the design of steel buildings yet. The concept of replacing a MDOF building where the masses and stiffnesses are in series by an equivalent SDOF system, presented by Calvi and Kingsley [11] for bridge structures where they act in parallel, also introduces the following issues.

1. Selection of an appropriate displaced shape and its effect on the design base shear, the static lateral force profile, and the member design requirements.
2. The effect of axial deformation of the columns on the displaced shape.
3. The greater cumulative effect of the gravity loads on the building ($P - \Delta$ effect).
4. The necessity of controlling the interstorey drift (storey ductility demand), particularly in the upper storeys, due to the influence of higher modes of vibration.

In the accompanying paper [5], this method is applied to the design of a two-storey and an eight-storey building with concentrically braced steel frames as the lateral load resisting system.

Acknowledgements

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